八十八年第一次專門職業及技術人員律師、會計師、建築師、技師、社會工作師、土地登記專業代理人檢覈筆試試題　**代號：1010**

類　　科：冷凍空調工程技師
科　　目：冷凍空調設計
考試時間：二小時

座號：＿＿＿＿＿

※注意：㈠本試題可使用電子計算器，但需詳列解答過程。
㈡不必抄題，作答時請將試題題號及答案依照順序寫在試卷上，於本試題上作答者，不予計分。

一、某房間原來的室內與室外溫度均爲36℃，相對濕度亦均爲90%。此房間的室內尺寸爲3m×4m×5m，其牆壁、地板與天花板能完全隔熱。若此房間使用一台窗型冷氣機，其壓縮機的熱效率爲90%，而房間內使用兩個功率均爲60W的燈泡，且希望在0.5小時內將室內空氣調節到25℃與60%相對濕度，則其壓縮機所需要的功率爲何？假設室內空氣隨時保持完全混合狀態，請做必要的合理假設。　（30分）

二、某小型冷凍系統使用一支毛細管做爲節流元件，若此毛細管的上、下游壓力分別爲p_u與p_d，且系統的冷凝器使冷媒達到過冷狀態，試以冷媒的性質（T-s或p-h圖）說明如何決定該毛細管的長度。　（25分）

三、某吸收式冷凍系統以天然氣的燃燒做爲熱源。假設天然氣的燃燒溫度爲2200K，而該冷凍系統的實際工作溫度如下：發生器(Generator)100℃、吸收器(Absorber)30℃、冷凝器(Condenser)40℃、蒸發器(Evaporator)10℃，試求此冷凍系統的理想COP。(10分) 若先以一部組合熱效率爲40%的熱機-發電機系統利用天然氣發電，然後驅動一套COP＝2.2的蒸氣壓縮式冷凍系統，試比較兩種系統的優劣。(15分)

四、某冷凍系統的蒸發與冷凝溫度分別爲-10℃與40℃，此系統原來使用R-22爲冷媒，而要改用R-134a爲冷媒，則此系統必須做那些修正才能達到所要的冷凍能力？(20分)

八十八年第一次專門職業及技術人員 律師、會計師 建築師、技師 社會工作師 土地登記專業代理人 檢覈筆試試題 **代號：1010**

類　科：冷凍空調工程技師
科　目：冷凍空調設計

表一

Moist air: thermodynamic properties of saturated air at atmospheric pressure of 101.325 kPa

t, °C	Vapor pressure, kPa	Humidity ratio, kg/kg	Specific volume, m^3kg	Enthalpy, kJ/kg
-40	0.01283	0.000079	0.6597	-40.041
-35	0.02233	0.000138	0.6740	-34.868
-30	0.03798	0.000234	0.6884	-29.600
-25	0.06324	0.000390	0.7028	-24.187
-20	0.10318	0.000637	0.7173	-18.546
-18	0.12482	0.000771	0.7231	-16.203
-16	0.15056	0.000930	0.7290	-13.795
-14	0.18107	0.001119	0.7349	-11.314
-12	0.21716	0.001342	0.7409	-8.745
-10	0.25971	0.001606	0.7469	-6.073
-8	0.30975	0.001916	0.7529	-3.285
-6	0.36846	0.002280	0.7591	-0.360
-4	0.43716	0.002707	0.7653	2.724
-2	0.51735	0.003206	0.7716	5.991
0	0.61072	0.003788	0.7781	9.470
1	0.6566	0.00407	0.7813	11.200
2	0.7055	0.00438	0.7845	12.978
3	0.7575	0.00471	0.7878	14.807
4	0.8130	0.00505	0.7911	16.692
5	0.8719	0.00542	0.7944	18.634
6	0.9347	0.00582	0.7978	20.639
7	1.0013	0.00624	0.8012	22.708
8	1.0722	0.00668	0.8046	24.848
9	1.1474	0.00716	0.8081	27.059
10	1.2272	0.00766	0.8116	29.348
11	1.3119	0.00820	0.8152	31.716
12	1.4017	0.00876	0.8188	34.172
13	1.4969	0.00937	0.8225	36.719
14	1.5977	0.01001	0.8262	39.362
15	1.7044	0.01069	0.8300	42.105
16	1.8173	0.01141	0.8338	44.955
17	1.9367	0.01218	0.8377	47.918
18	2.0630	0.01299	0.8417	50.998
19	2.1964	0.01384	0.8457	54.205
20	2.3373	0.01475	0.8498	57.544
21	2.4861	0.01572	0.8540	61.021
22	2.6431	0.01674	0.8583	64.646
23	2.8086	0.01781	0.8626	68.425
24	2.9832	0.01896	0.8671	72.366
25	3.1671	0.02016	0.8716	76.481
26	3.3609	0.02144	0.8763	80.777
27	3.5649	0.02279	0.8811	85.263
28	3.7797	0.02422	0.8860	89.952
29	4.0055	0.02572	0.8910	94.851
30	4.2431	0.02732	0.8961	99.977
31	4.4928	0.02900	0.9014	105.337
32	4.7552	0.03078	0.9068	110.946
33	5.0308	0.03266	0.9124	116.819
34	5.3201	0.03464	0.9182	122.968
35	5.6237	0.03674	0.9241	129.411
36	5.9423	0.03895	0.9302	136.161
37	6.2764	0.04129	0.9365	143.239
38	6.6265	0.04376	0.9430	150.660
39	6.9935	0.04636	0.9497	158.445
40	7.3778	0.04911	0.9567	166.615
40	7.3778	0.04911	0.9567	166.615
41	7.7803	0.05202	0.9639	175.192
42	8.2016	0.05509	0.9713	184.200
43	8.6424	0.05833	0.9790	193.662
44	9.1036	0.06176	0.9871	203.610
45	9.5856	0.06537	0.9954	214.067
46	10.0896	0.06920	1.0040	225.068

表二

Water: properties of liquid and saturated vapor

t, °C	Saturation pressure, kPa	Specific volume, m^3/kg		Enthalpy, kJ/kg		Entropy, kJ/kg · K	
		Liquid	Vapor	Liquid	Vapor	Liquid	Vapor
0	0.6108	0.0010002	206.3	-0.04	2501.6	-0.0002	9.1577
2	0.7055	0.0010001	179.9	8.39	2505.2	0.0306	9.1047
4	0.8129	0.0010000	157.3	16.80	2508.9	0.0611	9.0526
6	0.9345	0.0010000	137.8	25.21	2512.6	0.0913	9.0015
8	1.0720	0.0010001	121.0	33.60	2516.2	0.1213	8.9513
10	1.2270	0.0010003	106.4	41.99	2519.9	0.1510	8.9020
12	1.4014	0.0010004	93.84	50.38	2523.6	0.1805	8.8536
14	1.5973	0.0010007	82.90	58.75	2527.2	0.2098	8.8060
16	1.8168	0.0010010	73.38	67.13	2530.9	0.2388	8.7593
18	2.062	0.0010013	65.09	75.50	2534.5	0.2677	8.7135
20	2.337	0.0010017	57.84	83.86	2538.2	0.2963	8.6684
22	2.642	0.0010022	51.49	92.23	2541.8	0.3247	8.6241
24	2.982	0.0010026	45.93	100.59	2545.5	0.3530	8.5806
26	3.360	0.0010032	41.03	108.95	2549.1	0.3810	8.5379
28	3.778	0.0010037	36.73	117.31	2552.7	0.4088	8.4959
30	4.241	0.0010043	32.93	125.66	2556.4	0.4365	8.4546
32	4.753	0.0010049	29.57	134.02	2560.0	0.4640	8.4140
34	5.318	0.0010056	26.60	142.38	2563.6	0.4913	8.3740
36	5.940	0.0010063	23.97	150.74	2567.2	0.5184	8.3348
38	6.624	0.0010070	21.63	159.09	2570.8	0.5453	8.2962
40	7.375	0.0010078	19.55	167.45	2574.4	0.5721	8.2583
42	8.198	0.0010086	17.69	175.31	2577.9	0.5987	8.2209
44	9.100	0.0010094	16.04	184.17	2581.5	0.6252	8.1842
46	10.086	0.0010103	14.56	192.53	2585.1	0.6514	8.1481

類　科：冷凍空調工程技師
科　目：冷凍空調設計

表三　**Refrigerant 22 (Chlorodifluoromethane) Properties of Saturated Liquid and Saturated Vapor**

Temp.,ª °C	Absolute Pressure, MPa	Density, kg/m³ Liquid	Volume, m³/kg Vapor	Enthalpy, kJ/kg Liquid	Enthalpy, kJ/kg Vapor	Entropy, kJ/(kg·K) Liquid	Entropy, kJ/(kg·K) Vapor	Specific Heat c_p, kJ/(kg·K) Liquid	Specific Heat c_p, kJ/(kg·K) Vapor	c_p/c_v Vapor	Velocity of Sound, m/s Liquid	Velocity of Sound, m/s Vapor	Viscosity, μPa·s Liquid	Viscosity, μPa·s Vapor	Thermal Cond, mW/(m·K) Liquid	Thermal Cond, mW/(m·K) Vapor	Surface Tension, mN/m	Temp.,ª °C
−150.00	–	1701.5	–	26.01	335.85	0.0566	2.5752	–	0.434	1.285	–	123.	–	–	–	–	37.59	−150.00
−140.00	–	1675.3	–	43.84	340.24	0.1961	2.4222	–	0.445	1.275	–	128.	–	–	–	–	35.70	−140.00
−130.00	0.00006	1649.7	229.29	57.00	344.75	0.2916	2.3017	–	0.458	1.266	–	132.	–	–	–	–	33.84	−130.00
−120.00	0.00023	1624.0	63.648	68.51	349.38	0.3694	2.2033	–	0.470	1.258	–	136.	–	–	–	–	32.00	−120.00
−110.00	0.00074	1598.0	21.311	79.47	354.11	0.4386	2.1220	–	0.483	1.250	–	140.	–	–	–	–	30.17	−110.00
−100.00	0.00200	1571.7	8.2980	90.24	358.93	0.5027	2.0545	–	0.497	1.243	–	144.	–	–	–	–	28.37	−100.00
−90.00	0.00480	1545.1	3.6548	100.95	363.82	0.5629	1.9982	1.070	0.511	1.237	1094.	147.	–	–	–	–	26.59	−90.00
−80.00	0.01035	1518.3	1.7816	111.66	368.75	0.6197	1.9508	1.070	0.527	1.233	1037.	150.	–	–	–	–	24.83	−80.00
−70.00	0.02044	1491.1	0.94476	122.36	373.68	0.6738	1.9109	1.072	0.544	1.231	986.	153.	–	–	128.0	–	23.10	−70.00
−60.00	0.03747	1463.6	0.53734	133.11	378.58	0.7253	1.8770	1.076	0.563	1.231	937.	156.	–	–	123.1	5.61	21.39	−60.00
−50.00	0.06449	1435.5	0.32405	143.91	383.39	0.7748	1.8480	1.083	0.584	1.233	890.	158.	–	–	118.4	6.31	19.70	−50.00
−48.00	0.07140	1429.8	0.29469	146.08	384.35	0.7844	1.8427	1.085	0.589	1.233	881.	159.	–	–	117.5	6.44	19.37	−48.00
−46.00	0.07890	1424.1	0.26849	148.25	385.29	0.7940	1.8376	1.087	0.594	1.234	871.	159.	–	–	116.5	6.58	19.04	−46.00
−44.00	0.08700	1418.4	0.24507	150.43	386.23	0.8035	1.8326	1.089	0.598	1.235	862.	160.	–	–	115.6	6.71	18.70	−44.00
−42.00	0.09575	1412.6	0.22410	152.61	387.17	0.8130	1.8277	1.091	0.603	1.236	853.	160.	–	–	114.7	6.85	18.37	−42.00
−40.80b	0.10132	1409.1	0.21256	153.93	387.72	0.8186	1.8249	1.092	0.606	1.237	847.	160.	–	–	114.1	6.93	18.18	−40.80
−40.00	0.10518	1406.8	0.20526	154.80	388.09	0.8224	1.8230	1.093	0.608	1.237	844.	160.	–	–	113.8	6.98	18.05	−40.00
−38.00	0.11533	1401.0	0.18832	156.99	389.01	0.8317	1.8184	1.096	0.614	1.239	834.	161.	–	–	112.9	7.11	17.72	−38.00
−36.00	0.12623	1395.1	0.17306	159.19	389.93	0.8410	1.8140	1.098	0.619	1.240	825.	161.	–	–	112.0	7.24	17.39	−36.00
−34.00	0.13793	1389.2	0.15927	161.40	390.84	0.8502	1.8096	1.101	0.624	1.242	816.	161.	–	–	111.1	7.37	17.07	−34.00
−32.00	0.15045	1383.3	0.14680	163.61	391.74	0.8594	1.8054	1.104	0.630	1.243	807.	161.	–	–	110.1	7.50	16.74	−32.00
−30.00	0.16384	1377.3	0.13551	165.82	392.63	0.8685	1.8013	1.107	0.636	1.245	797.	162.	–	–	109.2	7.63	16.42	−30.00
−28.00	0.17815	1371.3	0.12525	168.04	393.52	0.8776	1.7973	1.110	0.642	1.247	788.	162.	–	–	108.4	7.76	16.10	−28.00
−26.00	0.19340	1365.2	0.11593	170.27	394.39	0.8866	1.7934	1.114	0.648	1.249	779.	162.	–	–	107.5	7.89	15.77	−26.00
−24.00	0.20965	1359.1	0.10744	172.51	395.26	0.8955	1.7896	1.117	0.654	1.252	770.	162.	–	–	106.6	8.02	–	−24.00
−22.00	0.22693	1352.9	0.09970	174.75	396.12	0.9044	1.7859	1.121	0.660	1.254	760.	163.	–	–	105.7	8.14	–	−22.00
−20.00	0.24529	1346.8	0.09262	177.00	396.97	0.9133	1.7822	1.125	0.667	1.257	751.	163.	260.1	–	104.8	8.27	–	−20.00
−18.00	0.26477	1340.5	0.08615	179.26	397.81	0.9222	1.7787	1.129	0.674	1.260	742.	163.	254.7	11.08	103.9	8.40	–	−18.00
−16.00	0.28542	1334.2	0.08023	181.53	398.64	0.9309	1.7752	1.133	0.681	1.263	733.	163.	249.4	11.16	103.1	8.52	–	−16.00
−14.00	0.30728	1327.9	0.07479	183.81	399.46	0.9397	1.7719	1.137	0.688	1.266	723.	163.	244.2	11.24	102.2	8.64	–	−14.00
−12.00	0.33040	1321.5	0.06979	186.09	400.27	0.9484	1.7686	1.141	0.695	1.269	714.	163.	239.1	11.32	101.3	8.77	–	−12.00
−10.00	0.35482	1315.0	0.06520	188.38	401.07	0.9571	1.7653	1.146	0.703	1.273	705.	163.	234.1	11.40	100.4	8.89	–	−10.00
−8.00	0.38059	1308.5	0.06096	190.69	401.85	0.9657	1.7621	1.151	0.710	1.277	696.	163.	229.1	11.48	99.6	9.02	–	−8.00
−6.00	0.40775	1301.9	0.05706	193.00	402.63	0.9743	1.7590	1.156	0.718	1.281	686.	163.	224.2	11.56	98.7	9.14	–	−6.00
−4.00	0.43636	1295.3	0.05345	195.32	403.39	0.9829	1.7560	1.161	0.727	1.285	677.	163.	219.4	11.64	97.9	9.26	–	−4.00
−2.00	0.46646	1288.6	0.05012	197.66	404.14	0.9915	1.7530	1.166	0.735	1.289	668.	163.	214.7	11.72	97.0	9.38	–	−2.00
0.00	0.49811	1281.8	0.04703	200.00	404.87	1.0000	1.7500	1.171	0.744	1.294	658.	163.	210.1	11.80	96.2	9.50	–	0.00
2.00	0.53134	1275.0	0.04417	202.35	405.59	1.0085	1.7471	1.177	0.753	1.299	649.	163.	205.6	11.88	95.3	9.63	–	2.00
4.00	0.56622	1268.1	0.04152	204.72	406.30	1.0170	1.7443	1.183	0.762	1.305	640.	163.	201.2	11.96	94.5	9.75	–	4.00
6.00	0.60279	1261.1	0.03906	207.10	406.99	1.0254	1.7415	1.189	0.772	1.310	630.	163.	196.9	12.04	93.6	9.87	–	6.00
8.00	0.64109	1254.0	0.03676	209.49	407.67	1.0338	1.7387	1.195	0.782	1.316	621.	163.	192.6	12.12	92.8	9.99	–	8.00
10.00	0.68119	1246.9	0.03463	211.89	408.33	1.0422	1.7360	1.202	0.792	1.323	611.	163.	188.5	12.20	92.0	10.11	–	10.00
12.00	0.72314	1239.7	0.03265	214.31	408.97	1.0506	1.7333	1.208	0.802	1.330	602.	162.	184.4	12.28	91.1	10.23	–	12.00
14.00	0.76698	1232.4	0.03079	216.74	409.60	1.0590	1.7306	1.215	0.813	1.337	592.	162.	180.5	12.36	90.3	10.35	–	14.00
16.00	0.81277	1225.0	0.02906	219.18	410.21	1.0673	1.7280	1.223	0.825	1.345	583.	162.	176.6	12.44	89.5	10.47	–	16.00
18.00	0.86056	1217.6	0.02744	221.63	410.80	1.0756	1.7254	1.230	0.837	1.353	573.	162.	172.8	12.52	88.7	10.59	–	18.00
20.00	0.91041	1210.0	0.02593	224.10	411.38	1.0840	1.7228	1.238	0.849	1.361	564.	161.	169.1	–	87.8	10.71	–	20.00
22.00	0.96236	1202.4	0.02451	226.59	411.93	1.0923	1.7202	1.246	0.862	1.370	554.	161.	165.4	–	87.0	10.82	–	22.00
24.00	1.0165	1194.6	0.02319	229.09	412.46	1.1006	1.7177	1.254	0.875	1.380	544.	160.	161.9	–	86.2	10.94	–	24.00
26.00	1.0728	1186.8	0.02194	231.60	412.98	1.1088	1.7151	1.263	0.889	1.391	535.	160.	158.4	–	85.4	11.06	–	26.00
28.00	1.1314	1178.8	0.02077	234.14	413.46	1.1171	1.7126	1.272	0.904	1.402	525.	160.	155.0	–	84.6	11.18	–	28.00
30.00	1.1924	1170.7	0.01968	236.69	413.93	1.1254	1.7101	1.282	0.919	1.413	515.	159.	151.7	–	83.8	11.30	–	30.00
32.00	1.2557	1162.5	0.01864	239.25	414.37	1.1336	1.7075	1.292	0.935	1.426	506.	159.	148.5	–	83.0	11.42	–	32.00
34.00	1.3215	1154.2	0.01767	241.84	414.79	1.1419	1.7050	1.302	0.952	1.440	496.	158.	145.4	–	82.2	11.54	–	34.00
36.00	1.3898	1145.7	0.01675	244.44	415.18	1.1501	1.7024	1.313	0.970	1.454	486.	158.	142.3	–	81.4	11.66	–	36.00
38.00	1.4606	1137.1	0.01589	247.06	415.54	1.1584	1.6999	1.325	0.989	1.470	476.	157.	139.3	–	80.6	11.78	–	38.00
40.00	1.5341	1128.4	0.01507	249.71	415.87	1.1667	1.6975	1.338	1.009	1.486	466.	156.	136.3	–	79.8	11.90	–	40.00
42.00	1.6103	1119.5	0.01430	252.37	416.17	1.1749	1.6947	1.351	1.030	1.504	456.	156.	–	–	79.0	12.02	–	42.00
44.00	1.6892	1110.4	0.01357	255.06	416.44	1.1832	1.6921	1.365	1.052	1.524	446.	155.	–	–	78.2	12.14	–	44.00
46.00	1.7709	1101.2	0.01288	257.77	416.68	1.1915	1.6894	1.380	1.076	1.545	436.	154.	–	–	77.4	12.26	–	46.00
48.00	1.8555	1091.8	0.01223	260.51	416.87	1.1998	1.6867	1.396	1.102	1.568	426.	153.	–	–	76.6	12.38	–	48.00
50.00	1.9431	1082.1	0.01161	263.27	417.03	1.2081	1.6840	1.414	1.129	1.593	415.	153.	–	–	–	–	–	50.00
55.00	2.1753	1057.1	0.01020	270.31	417.24	1.2291	1.6768	1.464	1.209	1.667	389.	150.	–	–	–	–	–	55.00
60.00	2.4274	1030.5	0.00895	277.56	417.14	1.2503	1.6692	1.528	1.307	1.762	362.	148.	–	–	–	–	–	60.00
65.00	2.7008	1001.8	0.00784	285.06	416.65	1.2718	1.6610	1.611	1.435	1.888	335.	145.	–	–	–	–	–	65.00
70.00	2.9967	970.4	0.00684	292.90	415.69	1.2940	1.6518	1.726	1.609	2.064	306.	142.	–	–	–	–	–	70.00
75.00	3.3168	935.3	0.00594	301.18	414.09	1.3169	1.6413	1.896	1.862	2.324	276.	138.	–	–	–	–	–	75.00
80.00	3.6627	894.8	0.00511	310.10	411.60	1.3413	1.6287	2.176	2.268	2.748	244.	134.	–	–	–	–	–	80.00
85.00	4.0368	845.1	0.00433	320.05	407.72	1.3680	1.6128	2.725	3.039	3.555	209.	129.	–	–	–	–	–	85.00
90.00	4.4416	777.5	0.00355	331.98	401.33	1.3998	1.5907	4.186	5.041	5.643	170.	124.	–	–	–	–	–	90.00
95.00	4.8820	665.4	0.00264	348.86	387.46	1.4442	1.5491	–	–	–	–	–	–	–	–	–	–	95.00
96.14c	4.9900	523.8	0.00191	366.59	366.59	1.4918	1.4918	∞	∞	∞	0.	0.	–	–	∞	∞	0.00	96.14

ªTemperatures are on the ITS–90 scale　　b = normal boiling point　　c = critical point

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八十八年第一次專門職業及技術人員社會工作師檢覈筆試試題 **代號：1010**
土地登記專業代理人

類　科：冷凍空調工程技師
科　目：冷凍空調設計

表四

Refrigerant 134a (1,1,1,2-Tetrafluoroethane) Properties of Saturated Liquid and Saturated Vapor

Temp.,* °C	Absolute Pressure, MPa	Density, kg/m³ Liquid	Volume, m³/kg Vapor	Enthalpy, kJ/kg Liquid	Enthalpy, kJ/kg Vapor	Entropy, kJ/(kg·K) Liquid	Entropy, kJ/(kg·K) Vapor	Specific Heat c_p, kJ/(kg·K) Liquid	Specific Heat c_p, kJ/(kg·K) Vapor	c_p/c_v, Vapor	Velocity of Sound, m/s Liquid	Velocity of Sound, m/s Vapor	Viscosity, μPa·s Liquid	Viscosity, μPa·s Vapor	Thermal Cond, mW/(m·K) Liquid	Thermal Cond, mW/(m·K) Vapor	Surface Tension, mN/m	Temp.,* °C
−103.30a	0.00039	1591.2	35.263	71.89	335.07	0.4143	1.9638	1.147	0.585	1.163	1135.	127.	2186.6	6.63	–	–	28.15	−103.30
−100.00	0.00056	1581.9	25.039	75.71	337.00	0.4366	1.9456	1.168	0.592	1.161	1111.	128.	1958.2	6.76	–	–	27.56	−100.00
−90.00	0.00153	1553.9	9.7191	87.59	342.94	0.5032	1.8975	1.201	0.614	1.155	1051.	131.	1445.6	7.16	–	–	25.81	−90.00
−80.00	0.00369	1526.2	4.2504	99.65	349.03	0.5674	1.8585	1.211	0.637	1.151	999.	134.	1109.9	7.57	–	–	24.11	−80.00
−70.00	0.00801	1498.6	2.0528	111.78	355.23	0.6286	1.8269	1.215	0.660	1.148	951.	137.	879.6	7.97	125.8	–	22.44	−70.00
−60.00	0.01594	1471.0	1.0770	123.96	361.51	0.6871	1.8016	1.220	0.685	1.146	904.	139.	715.4	8.38	121.1	–	20.81	−60.00
−50.00	0.02948	1443.1	0.60560	136.21	367.83	0.7432	1.7812	1.229	0.712	1.146	858.	142.	594.3	8.79	116.5	7.12	19.22	−50.00
−40.00	0.05122	1414.8	0.36095	148.57	374.16	0.7973	1.7649	1.243	0.740	1.148	812.	144.	502.2	9.20	111.9	8.19	17.66	−40.00
−30.00	0.08436	1385.9	0.22596	161.10	380.45	0.8498	1.7519	1.260	0.771	1.152	765.	145.	430.4	9.62	107.3	9.16	16.13	−30.00
−28.00	0.09268	1380.0	0.20682	163.62	381.70	0.8601	1.7497	1.264	0.778	1.153	756.	145.	418.0	9.71	106.3	9.35	15.83	−28.00
−26.07b	0.10132	1374.3	0.19016	166.07	382.90	0.8701	1.7476	1.268	0.784	1.154	747.	146.	406.4	9.79	105.4	9.52	15.54	−26.07
−26.00	0.10164	1374.1	0.18961	166.16	382.94	0.8704	1.7476	1.268	0.785	1.154	747.	146.	406.0	9.79	105.4	9.53	15.53	−26.00
−24.00	0.11127	1368.2	0.17410	168.70	384.19	0.8806	1.7455	1.273	0.791	1.155	738.	146.	394.6	9.88	104.5	9.71	15.23	−24.00
−22.00	0.12160	1362.2	0.16010	171.26	385.43	0.8908	1.7436	1.277	0.798	1.156	728.	146.	383.6	9.96	103.6	9.89	14.93	−22.00
−20.00	0.13268	1356.2	0.14744	173.82	386.66	0.9009	1.7417	1.282	0.805	1.157	719.	146.	373.1	10.05	102.6	10.07	14.63	−20.00
−18.00	0.14454	1350.2	0.13597	176.39	387.89	0.9110	1.7399	1.286	0.812	1.159	710.	146.	363.0	10.14	101.7	10.24	14.33	−18.00
−16.00	0.15721	1344.1	0.12556	178.97	389.11	0.9211	1.7383	1.291	0.820	1.160	700.	147.	353.3	10.22	100.8	10.42	14.04	−16.00
−14.00	0.17074	1338.0	0.11610	181.56	390.33	0.9311	1.7367	1.296	0.827	1.162	691.	147.	344.0	10.31	99.9	10.59	13.74	−14.00
−12.00	0.18516	1331.8	0.10749	184.16	391.55	0.9410	1.7351	1.301	0.835	1.164	682.	147.	335.0	10.40	99.0	10.76	13.45	−12.00
−10.00	0.20052	1325.6	0.09963	186.78	392.75	0.9509	1.7337	1.306	0.842	1.166	672.	147.	326.3	10.49	98.0	10.93	13.16	−10.00
−8.00	0.21684	1319.3	0.09246	189.40	393.95	0.9608	1.7323	1.312	0.850	1.168	663.	147.	318.0	10.58	97.1	11.10	12.87	−8.00
−6.00	0.23418	1313.0	0.08591	192.03	395.15	0.9707	1.7310	1.317	0.858	1.170	654.	147.	309.9	10.67	96.2	11.28	12.58	−6.00
−4.00	0.25257	1306.6	0.07991	194.68	396.33	0.9805	1.7297	1.323	0.866	1.172	644.	147.	302.2	10.76	95.3	11.45	12.29	−4.00
−2.00	0.27206	1300.2	0.07440	197.33	397.51	0.9903	1.7285	1.329	0.875	1.175	635.	147.	294.7	10.85	94.3	11.62	12.00	−2.00
0.00	0.29269	1293.7	0.06935	200.00	398.68	1.0000	1.7274	1.335	0.883	1.178	626.	147.	287.4	10.94	93.4	11.79	11.71	0.00
2.00	0.31450	1287.1	0.06470	202.68	399.84	1.0097	1.7263	1.341	0.892	1.180	616.	147.	280.4	11.03	92.5	11.96	11.43	2.00
4.00	0.33755	1280.5	0.06042	205.37	401.00	1.0194	1.7252	1.347	0.901	1.183	607.	147.	273.6	11.13	91.6	12.13	11.14	4.00
6.00	0.36186	1273.8	0.05648	208.08	402.14	1.0291	1.7242	1.353	0.910	1.187	598.	147.	267.0	11.22	90.7	12.31	10.86	6.00
8.00	0.38749	1267.0	0.05284	210.80	403.27	1.0387	1.7233	1.360	0.920	1.190	588.	147.	260.6	11.32	89.7	12.48	10.58	8.00
10.00	0.41449	1260.2	0.04948	213.53	404.40	1.0483	1.7224	1.367	0.930	1.193	579.	146.	254.3	11.42	88.8	12.66	10.30	10.00
12.00	0.44289	1253.3	0.04636	216.27	405.51	1.0579	1.7215	1.374	0.939	1.197	569.	146.	248.3	11.52	87.9	12.84	10.02	12.00
14.00	0.47276	1246.3	0.04348	219.03	406.61	1.0674	1.7207	1.381	0.950	1.201	560.	146.	242.5	11.62	87.0	13.02	9.74	14.00
16.00	0.50413	1239.3	0.04081	221.80	407.70	1.0770	1.7199	1.388	0.960	1.206	550.	146.	236.8	11.72	86.0	13.20	9.47	16.00
18.00	0.53706	1232.1	0.03833	224.59	408.78	1.0865	1.7191	1.396	0.971	1.210	541.	146.	231.2	11.82	85.1	13.39	9.19	18.00
20.00	0.57159	1224.9	0.03603	227.40	409.84	1.0960	1.7183	1.404	0.982	1.215	532.	145.	225.8	11.92	84.2	13.57	8.92	20.00
22.00	0.60777	1217.5	0.03388	230.21	410.89	1.1055	1.7176	1.412	0.994	1.220	522.	145.	220.5	12.03	83.3	13.76	8.65	22.00
24.00	0.64566	1210.1	0.03189	233.05	411.93	1.1149	1.7169	1.420	1.006	1.226	512.	145.	215.4	12.14	82.4	13.96	8.38	24.00
26.00	0.68531	1202.6	0.03003	235.90	412.95	1.1244	1.7162	1.429	1.018	1.231	503.	144.	210.4	12.25	81.4	14.15	8.11	26.00
28.00	0.72676	1194.9	0.02829	238.77	413.95	1.1338	1.7155	1.438	1.031	1.238	493.	144.	205.5	12.36	80.5	14.35	7.84	28.00
30.00	0.77008	1187.2	0.02667	241.65	414.94	1.1432	1.7149	1.447	1.044	1.244	484.	143.	200.7	12.48	79.6	14.56	7.57	30.00
32.00	0.81530	1179.3	0.02516	244.55	415.90	1.1527	1.7142	1.457	1.058	1.251	474.	143.	196.0	12.60	78.7	14.76	7.31	32.00
34.00	0.86250	1171.3	0.02374	247.47	416.85	1.1621	1.7135	1.467	1.073	1.259	465.	142.	191.4	12.72	77.7	14.97	7.05	34.00
36.00	0.91172	1163.2	0.02241	250.41	417.78	1.1715	1.7129	1.478	1.088	1.267	455.	142.	186.9	12.84	76.8	15.19	6.78	36.00
38.00	0.96301	1154.9	0.02116	253.37	418.69	1.1809	1.7122	1.489	1.104	1.276	445.	141.	182.5	12.97	75.9	15.41	6.52	38.00
40.00	1.0165	1146.5	0.01999	256.35	419.58	1.1903	1.7115	1.500	1.120	1.285	436.	140.	178.2	13.10	75.0	15.64	6.27	40.00
42.00	1.0721	1137.9	0.01890	259.35	420.44	1.1997	1.7108	1.513	1.138	1.295	426.	140.	174.0	13.24	74.1	15.86	6.01	42.00
44.00	1.1300	1129.2	0.01786	262.38	421.28	1.2091	1.7101	1.525	1.156	1.306	416.	139.	169.8	13.38	73.1	16.10	5.76	44.00
46.00	1.1901	1120.3	0.01689	265.42	422.09	1.2185	1.7094	1.539	1.175	1.318	407.	138.	165.7	13.52	72.2	16.34	5.51	46.00
48.00	1.2527	1111.3	0.01598	268.49	422.88	1.2279	1.7086	1.553	1.196	1.331	397.	137.	161.7	13.67	71.3	16.59	5.26	48.00
50.00	1.3177	1102.0	0.01511	271.59	423.63	1.2373	1.7078	1.569	1.218	1.345	387.	137.	157.7	13.83	70.4	16.84	5.01	50.00
52.00	1.3852	1092.6	0.01430	274.71	424.35	1.2468	1.7070	1.585	1.241	1.360	377.	136.	153.8	13.99	69.5	17.10	4.76	52.00
54.00	1.4553	1082.9	0.01353	277.86	425.03	1.2562	1.7061	1.602	1.266	1.377	367.	135.	149.9	14.16	68.5	17.36	4.52	54.00
56.00	1.5280	1073.0	0.01280	281.04	425.68	1.2657	1.7051	1.621	1.293	1.395	358.	134.	146.1	14.33	67.6	17.63	4.28	56.00
58.00	1.6053	1062.8	0.01212	284.25	426.29	1.2752	1.7041	1.641	1.322	1.416	348.	134.	142.3	14.51	66.7	17.91	4.04	58.00
60.00	1.6815	1052.4	0.01146	287.49	426.86	1.2847	1.7031	1.663	1.354	1.438	338.	132.	138.6	14.71	65.8	18.19	3.81	60.00
62.00	1.7625	1041.7	0.01085	290.77	427.37	1.2943	1.7019	1.686	1.388	1.463	328.	131.	134.9	14.91	64.9	18.48	3.57	62.00
64.00	1.8464	1030.7	0.01026	294.08	427.84	1.3039	1.7007	1.712	1.426	1.490	318.	129.	131.2	15.12	63.9	18.78	3.34	64.00
66.00	1.9334	1019.4	0.00970	297.44	428.25	1.3136	1.6993	1.740	1.468	1.522	308.	128.	127.5	15.35	63.0	19.09	3.12	66.00
68.00	2.0234	1007.7	0.00917	300.84	428.61	1.3234	1.6979	1.772	1.515	1.557	298.	127.	123.9	15.59	62.1	19.40	2.89	68.00
70.00	2.1165	995.6	0.00867	304.29	428.89	1.3332	1.6963	1.806	1.567	1.597	287.	126.	120.3	15.85	61.2	19.72	2.67	70.00
72.00	2.2130	983.1	0.00818	307.79	429.10	1.3430	1.6945	1.846	1.626	1.642	277.	124.	116.7	16.12	60.3	20.05	2.46	72.00
74.00	2.3127	970.0	0.00772	311.34	429.23	1.3530	1.6926	1.890	1.693	1.695	267.	123.	113.1	16.41	59.3	20.39	2.24	74.00
76.00	2.4159	956.5	0.00728	314.96	429.27	1.3631	1.6905	1.941	1.770	1.757	256.	121.	109.4	16.73	58.4	20.74	2.03	76.00
78.00	2.5227	942.3	0.00686	318.65	429.20	1.3733	1.6881	2.000	1.861	1.830	246.	120.	105.8	17.08	57.5	21.09	1.83	78.00
80.00	2.6331	927.4	0.00646	322.41	429.02	1.3837	1.6855	2.069	1.967	1.917	235.	118.	102.1	17.46	56.6	21.46	1.63	80.00
85.00	2.9259	886.2	0.00550	332.27	427.91	1.4105	1.6775	2.313	2.348	2.231	207.	113.	92.7	18.59	54.3	22.41	1.15	85.00
90.00	3.2445	836.9	0.00461	343.01	425.48	1.4392	1.6663	2.766	3.064	2.832	178.	108.	82.6	20.15	–	–	0.72	90.00
95.00	3.5916	771.6	0.00374	355.43	420.60	1.4720	1.6490	3.961	4.942	4.424	145.	102.	70.9	22.59	–	–	0.33	95.00
100.00	3.9721	646.7	0.00265	374.02	407.08	1.5207	1.6093	–	–	–	105.	94.	53.0	28.86	–	–	0.03	100.00
101.03c	4.0560	513.3	0.00195	389.79	389.79	1.5593	1.5593	∞	∞	∞	0	0	–	–	∞	∞	0	101.03

*Temperatures are on the ITS–90 scale　　a = triple point　　b = normal boiling point　　c = critical point